

Backend PROJECTS

Save it for later !

20

Beginner Level

- **To-Do List API** : Full CRUD for managing tasks (Node.js + Express + MongoDB or SQLite).
- **Notes API**: Simple authentication + private notes per user.
- **Bookshelf API**: Manage books, authors, and categories.
- **URL Shortener**: Convert long URLs into short codes.
- **Weather Proxy API**: Consume external weather APIs (e.g., OpenWeatherMap) and expose simplified results.

Intermediate

- **Blog Platform API:** Users, posts, and comments (CRUD) with JWT authentication and pagination.
- **Movie Review API:** Allow users to rate and review movies; compute average ratings.
- **Task Management System:** Projects, users, and tasks with deadlines and assignments.
- **Chatroom REST API:** Create chatrooms and send stored messages; optional WebSocket upgrade.
- **Mini E-commerce Backend:** Manage products, carts, and orders with stock and price logic.

Advanced

- **Authentication Service:** Centralized token service with JWT or OAuth2, including refresh tokens.
- **Image Upload & Processing API:** Accept and resize images; store in S3 or similar.
- **Booking System API:** Reserve resources (e.g., rooms or slots), handle conflicts and cancellations.
- **Billing & Invoicing API:** Stripe integration, PDF invoice generation, and subscription tracking.
- **Analytics Tracker API:** Track events (page views, clicks) and display stats with caching.

Expert

- **Microservices E-commerce Platform:** Split services (cart, catalog, auth, etc.) with Kafka or RabbitMQ.
- **Real-Time Chat Server:** WebSocket chat app using Redis Pub/Sub for scaling.
- **Video Processing Backend:** Upload videos, compress and generate thumbnails using FFmpeg.
- **CI/CD Automation Server:** Git repo job runner with logs and status updates.
- **AI-Enhanced Search Engine:** Search engine with Elasticsearch + TF-IDF or vector-based ranking.